Lab Title:- STEGANOGRAPHY AND HIDDEN DATA DETECTION

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Course Name:- COMPUTER AND DIGITAL FORENSICS

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Executive summary

The lab covers a practical hands-on activity about steganography by embedding file into another file and then extracting that same file.

Lab Objectives

The objectives of this lab is to understand steganography and how important is steganography in digital forensic.

Tools and Resources Used

- Kali linux:- - A penetration testing platform used for security assessments.

Kali linux tools

- steghide:- A steganography program that is able to hide data in various kinds of image- and audio-files
- stegcracker:- A steganography brute-force utility to uncover hidden data inside files.
- stegseek:- used together with a wordlist to search for passphase.
- md5sum:- Print or check MD5 (128-bit) checksums.
- sha256sum:- Print or check SHA256 (256-bit) checksums.

Methology

- I launched the kali linux terminal
- I used the wget command to download the image file
- I obtained the md5/sha256 hash of the image file using the md5sum/sha256sum command
- I created a text file to be embedded into the image file
- I obtained the md5/sha256 hash of the text file created using the md5sum/sha256sum command

- I embedded the text file into the image file
- I obtained the md5/sha256 hash of the image file using the md5sum/sha256sum command after embedding
- I extract the text file from the image file
- I obtained the md5/sha256 hash of the text file using the md5sum/sha256sum command after extracting

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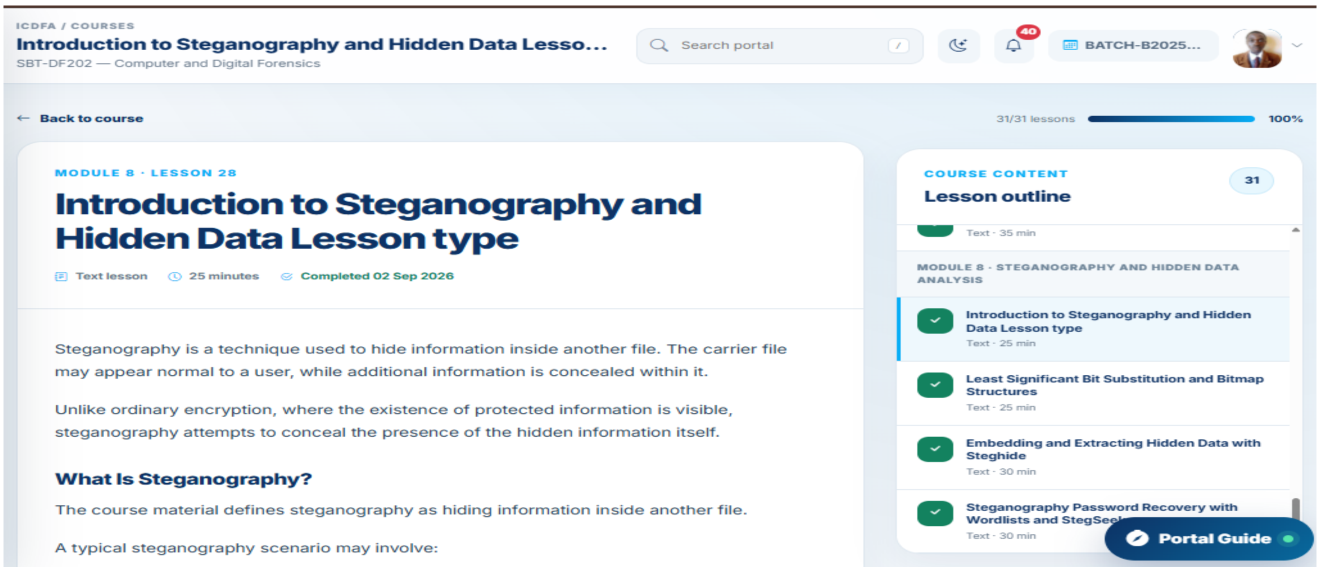
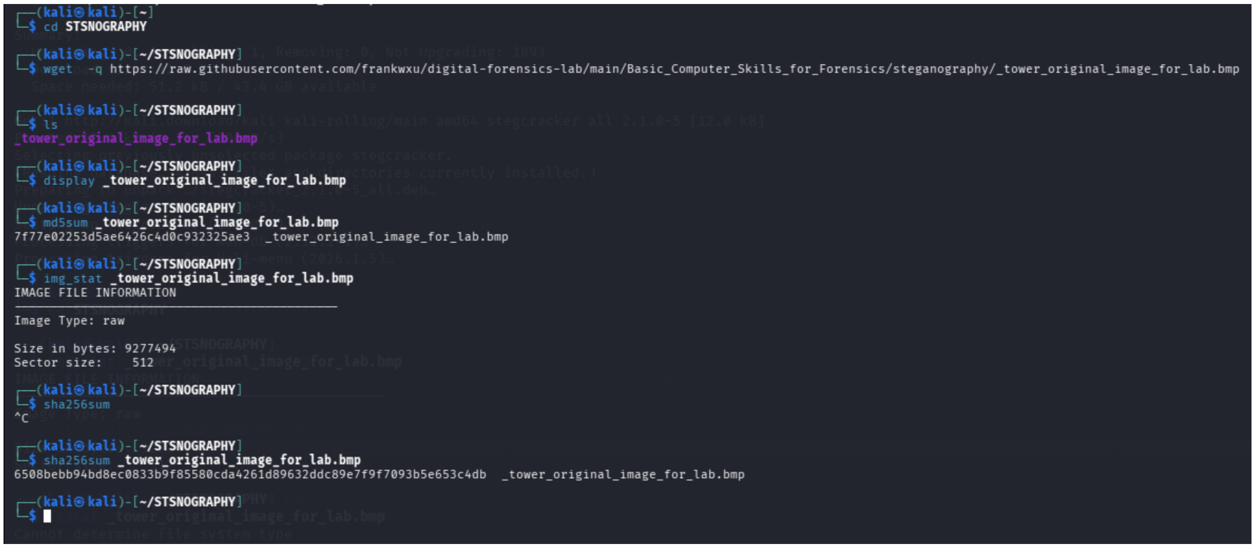
Screenshot and Evidence

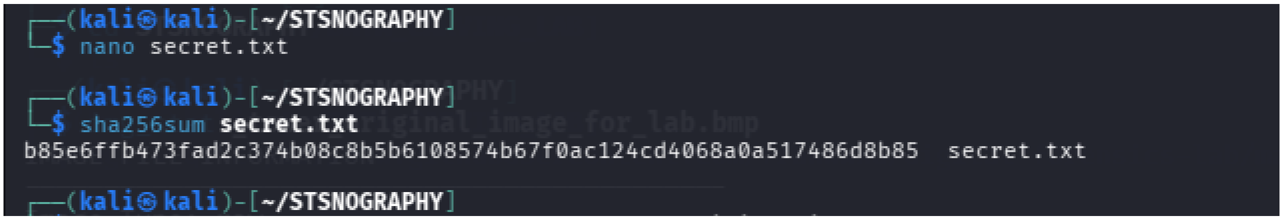
SCREENSHOT SHOWING MODULE 8 REVIEW



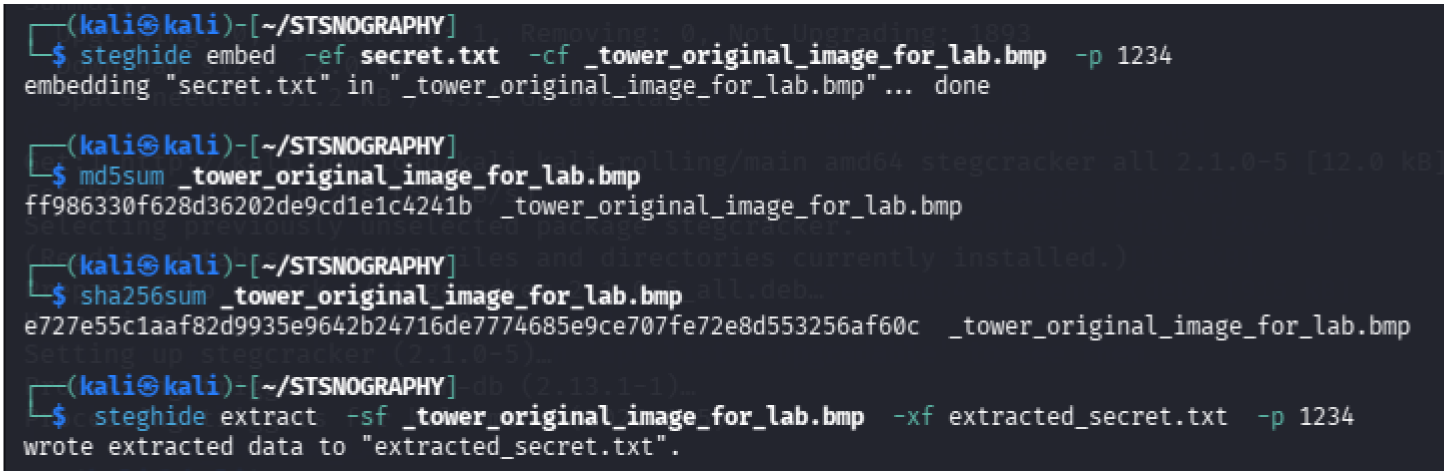
SCREENSHOT SHOWING PREPARATION OF THE LAB ENVIRONMENT & ORIGINAL CARRIER IMAGE AND ITS HASH



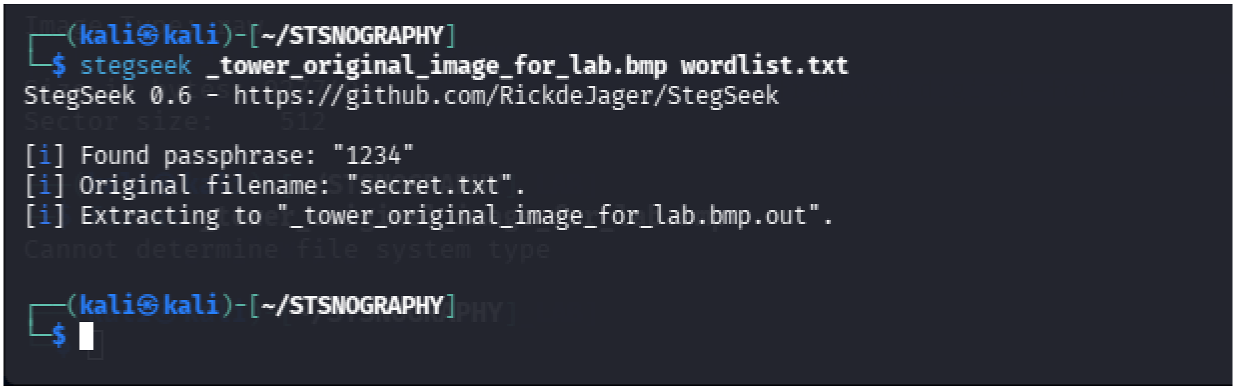
SCREENSHOT SHOWING SECRET EVIDENCE FILE WITH ITS HASH VALUE



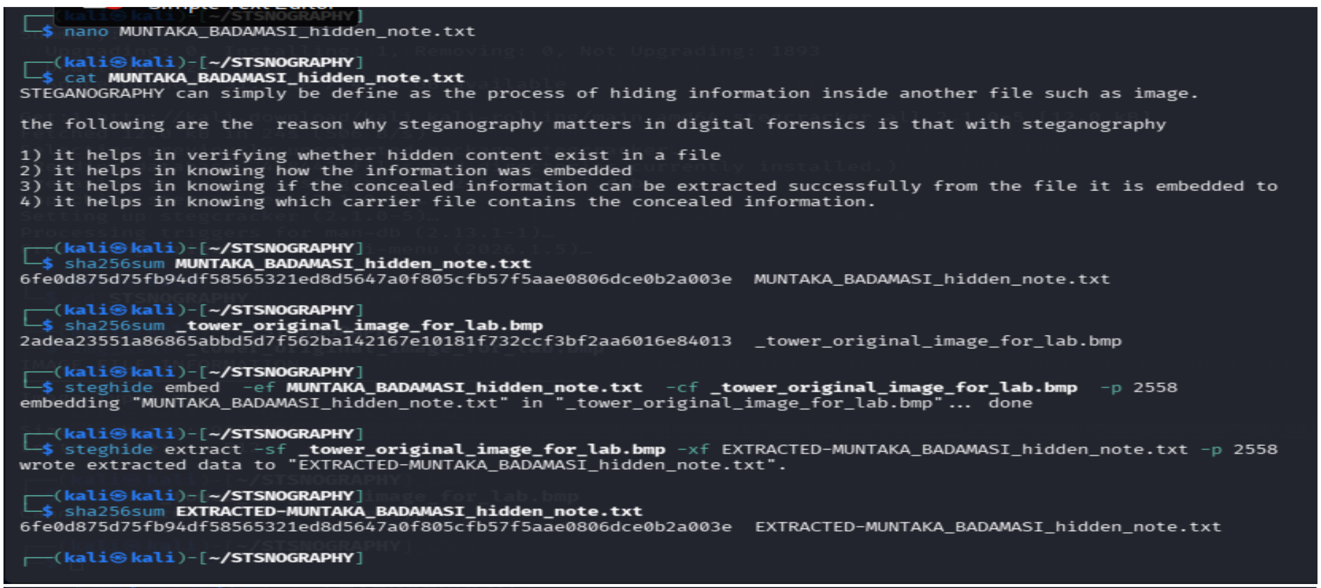
SCREENSHOT SHOWING SUCCESSFUL EMBEDDING & EXTRACTION OF EVIDENCE FILE FROM THE CARRIER IMAGE

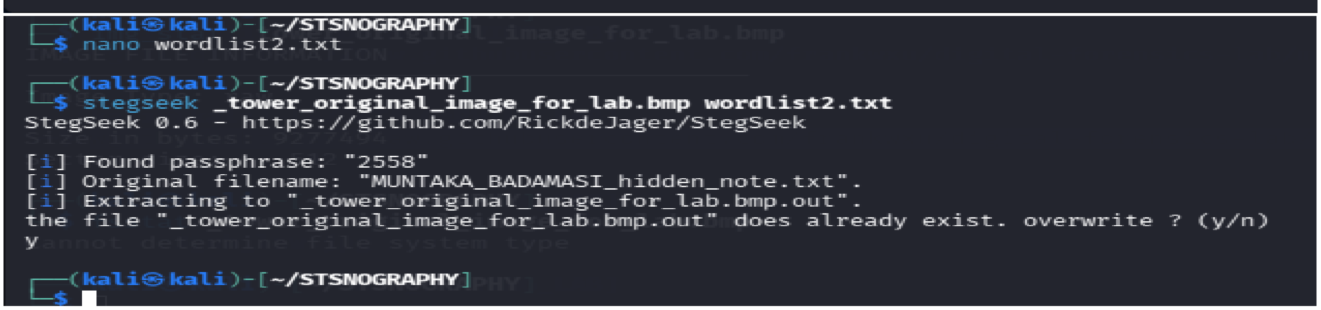


SCREENSHOT SHOWING CONTROLLED PASSWORD DICTIONARY TEST



SCREENSHOT SHOWING STUDENT-CREATED INDEPENDENT STEGANOGRAPHY EVIDENCE FILE TASK





Conclusion

In conclusion the lab has laid a foundation and also expose me to understanding steganography and also how important is steganography in digital forensic.

Reference

➢ICDFA course Material (BVDF201(computer and digital forensics)- by COMMANDANT AMINU IDRIS

➢ICDFA Recorded sessions